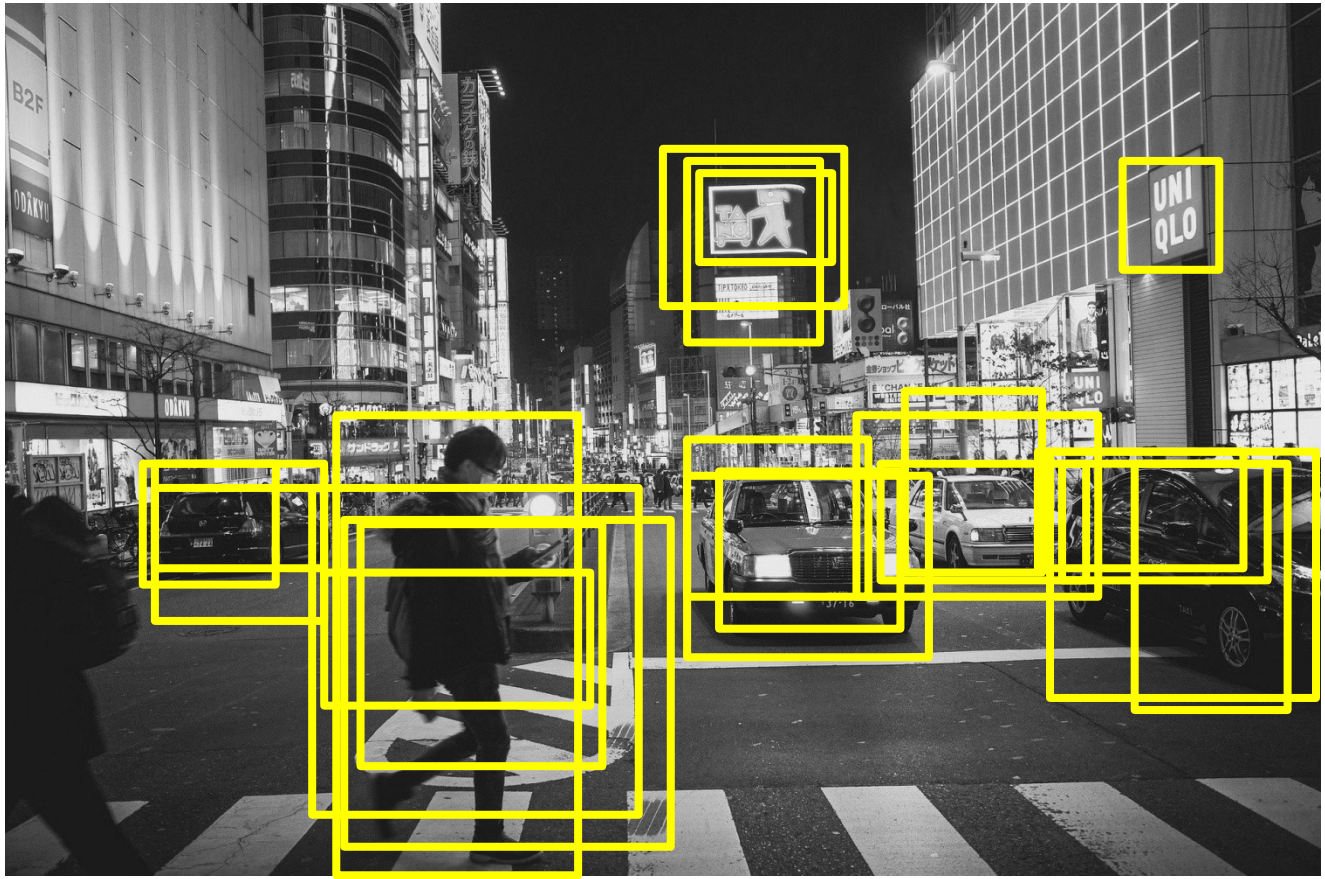
Computer Vision with Embedded Machine Learning

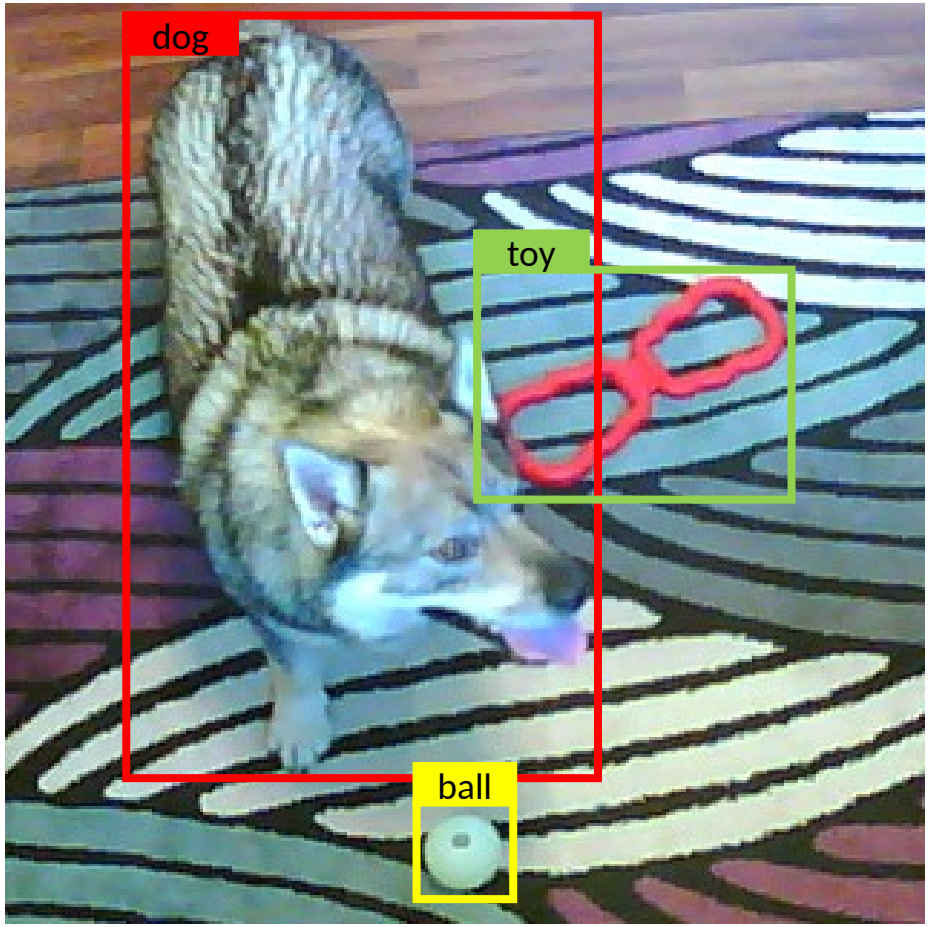
Object Detection Models

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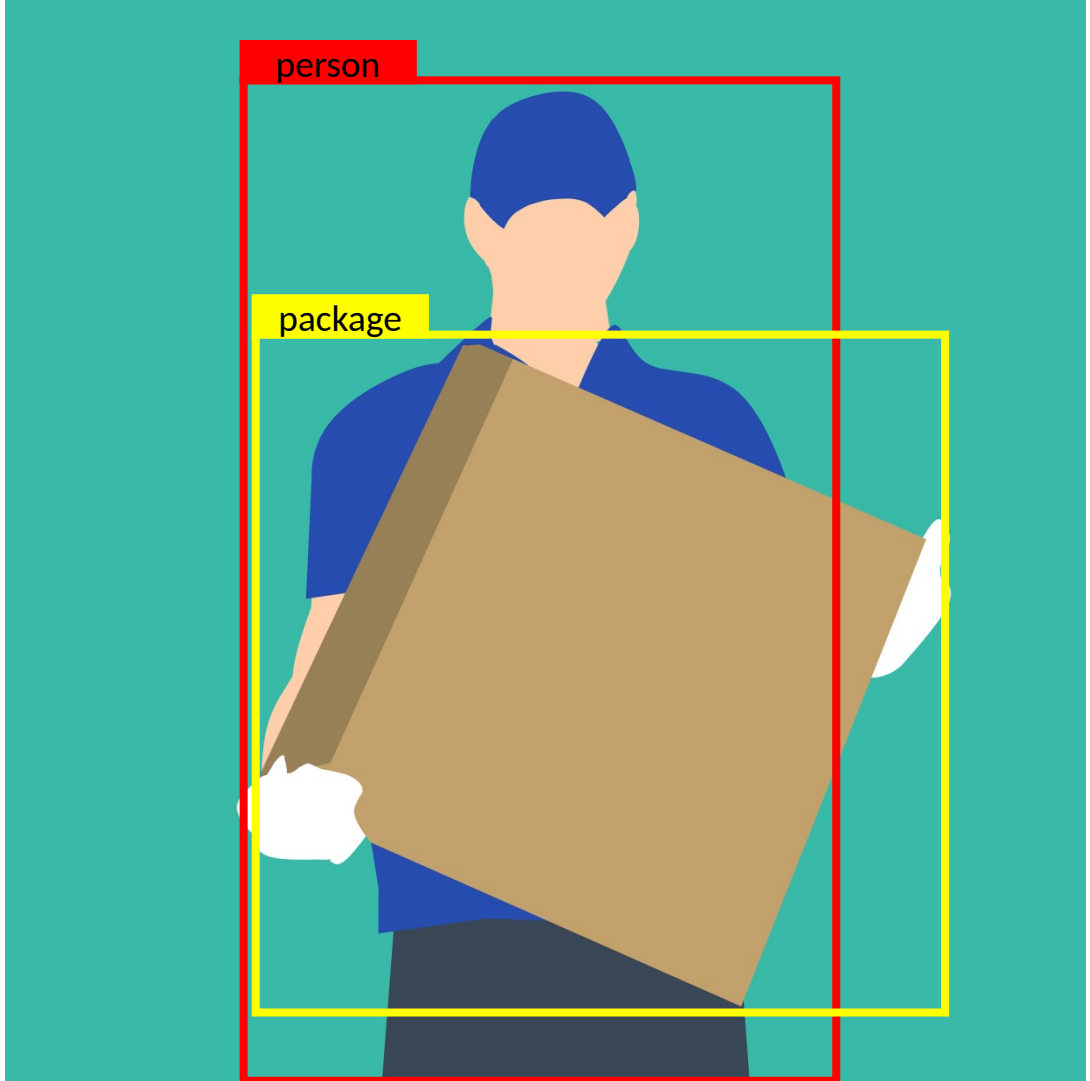
Region-based Convolutional Neural Network (R-CNN)



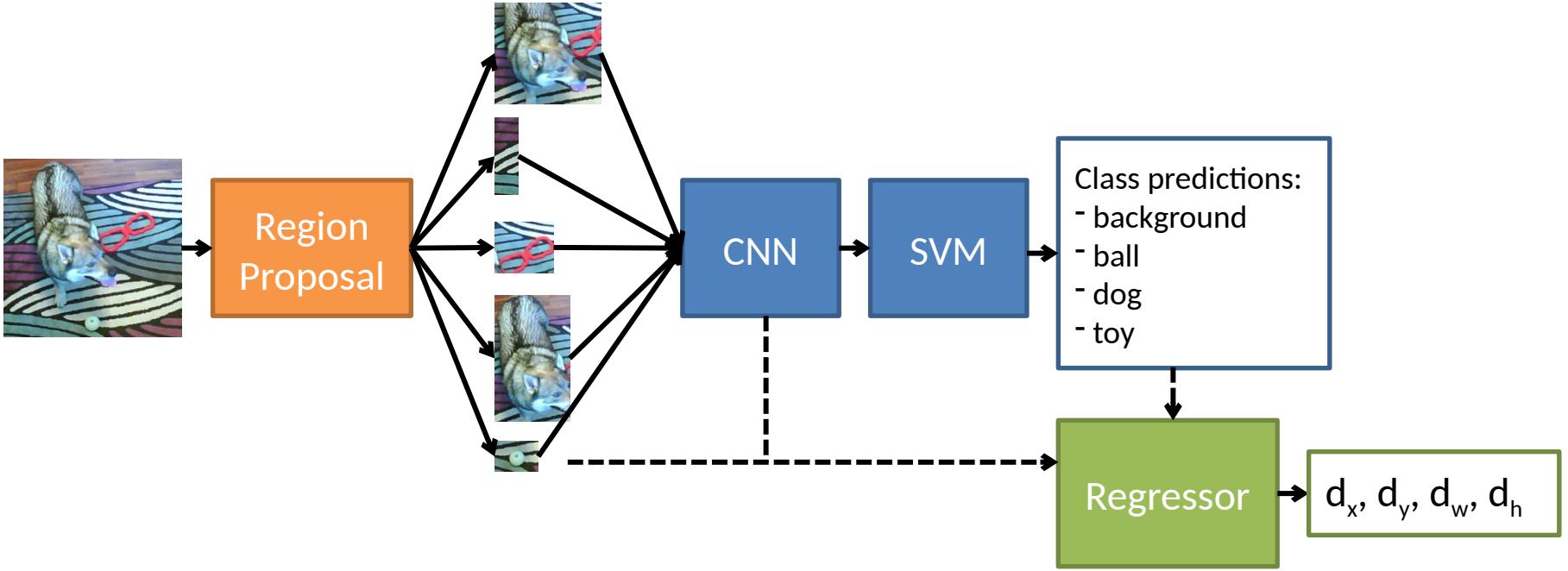
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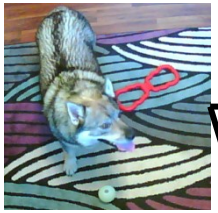

person
package

R-CNN



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Fast R-CNN



CNN

ROI Pooling

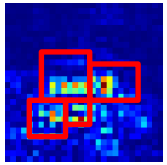
FC Layers

Flatten

FC Layer

Class predictions:
- background
- ball
- dog
- toy

Region Proposal

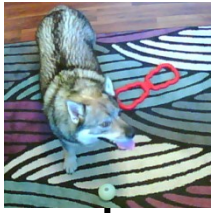


Regressor (FC Layer)

d_x, d_y, d_w, d_h

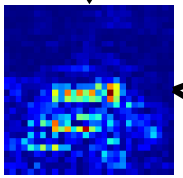
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Faster R-CNN



CNN

e.g. VGG19, MobileNet



ROI Pooling

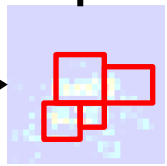
FC Layers

Flatten

FC Layer

Class predictions:
- background
- ball
- dog
- toy

Region Proposal Network

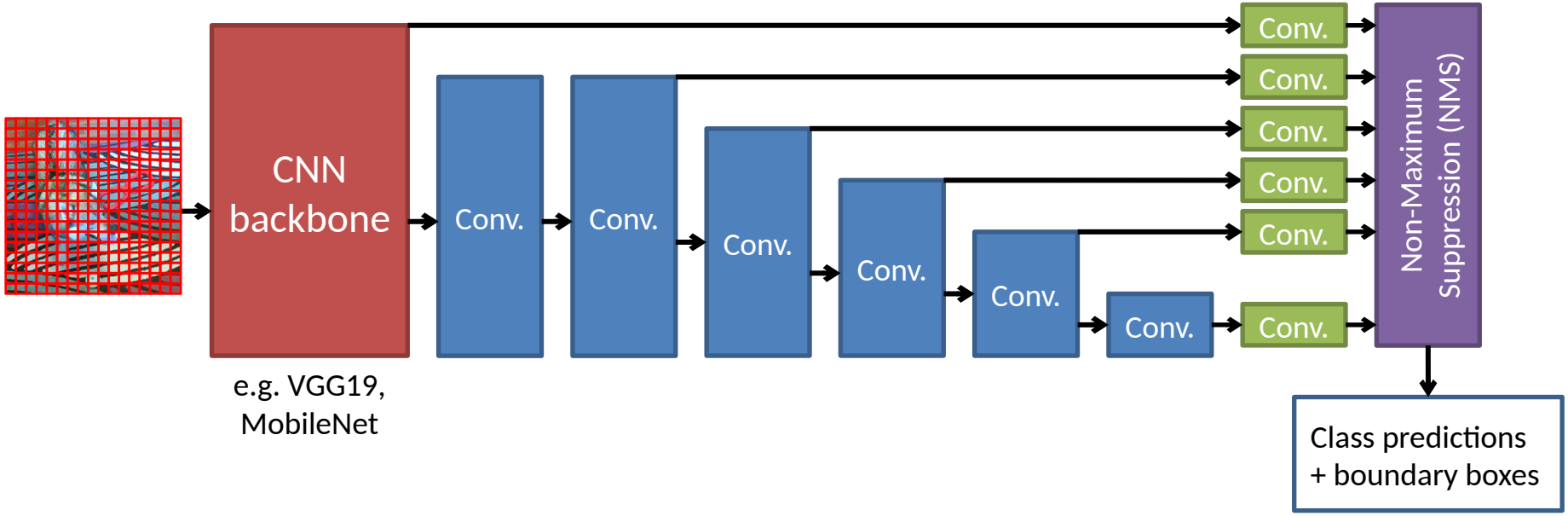


Regressor (FC Layer)

d_x, d_y, d_w, d_h

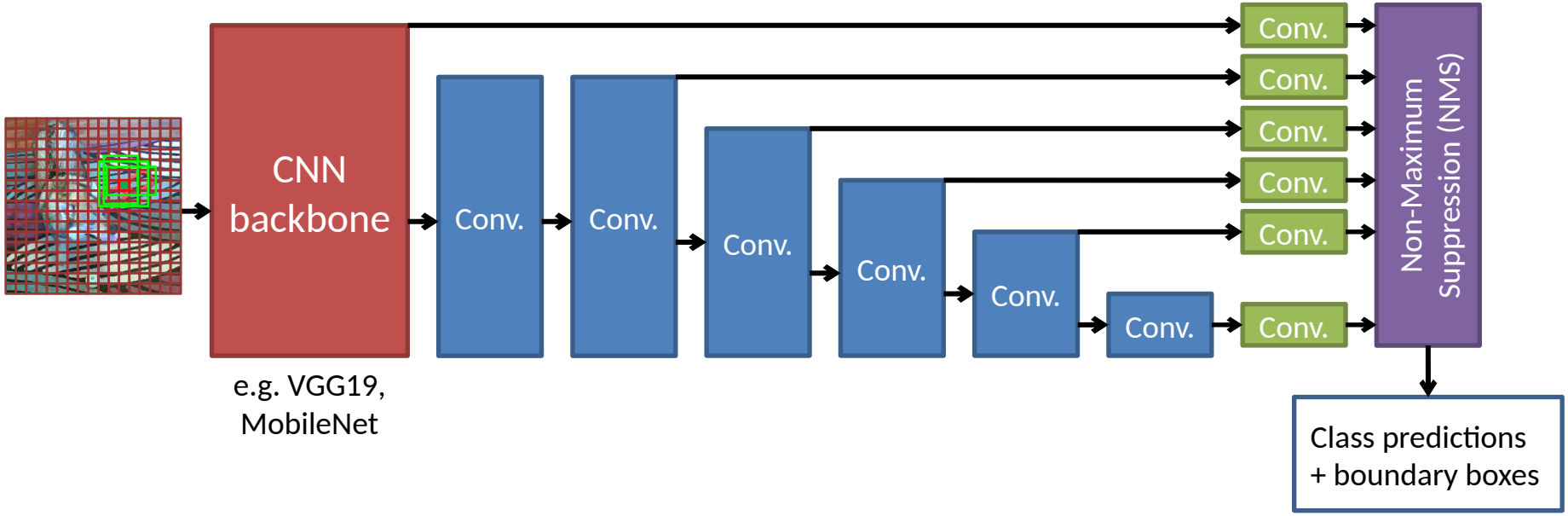
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Single Shot MultiBox Detector (SSD)



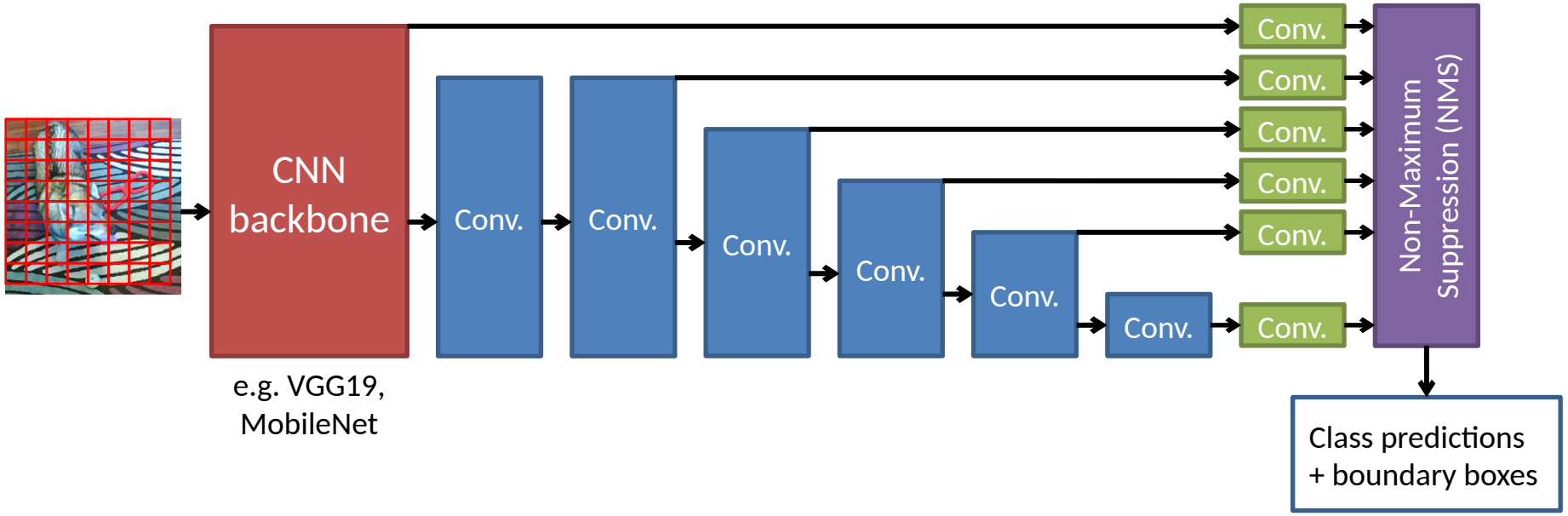
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Single Shot MultiBox Detector (SSD)



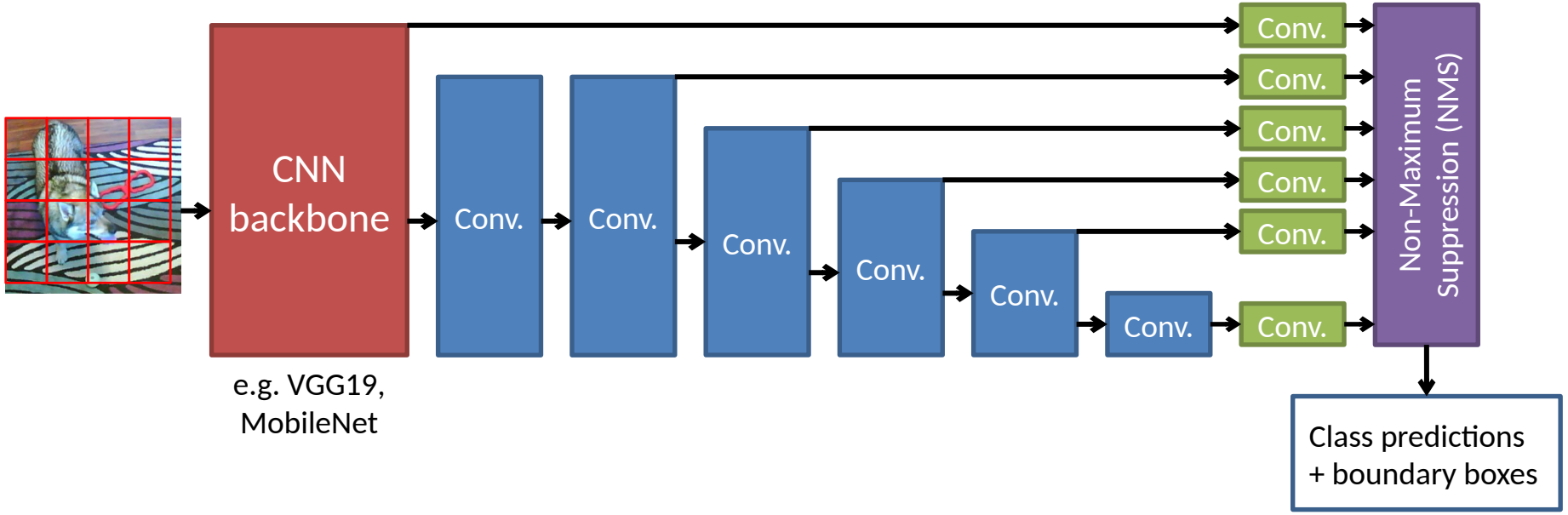
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Single Shot MultiBox Detector (SSD)



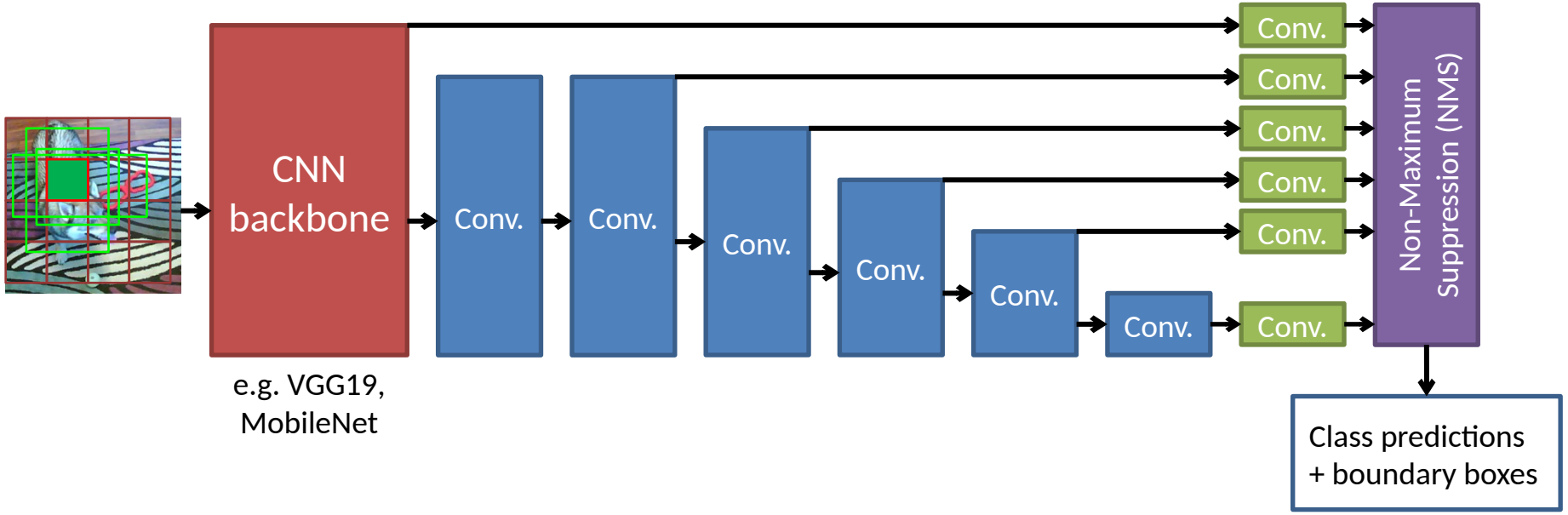
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Single Shot MultiBox Detector (SSD)



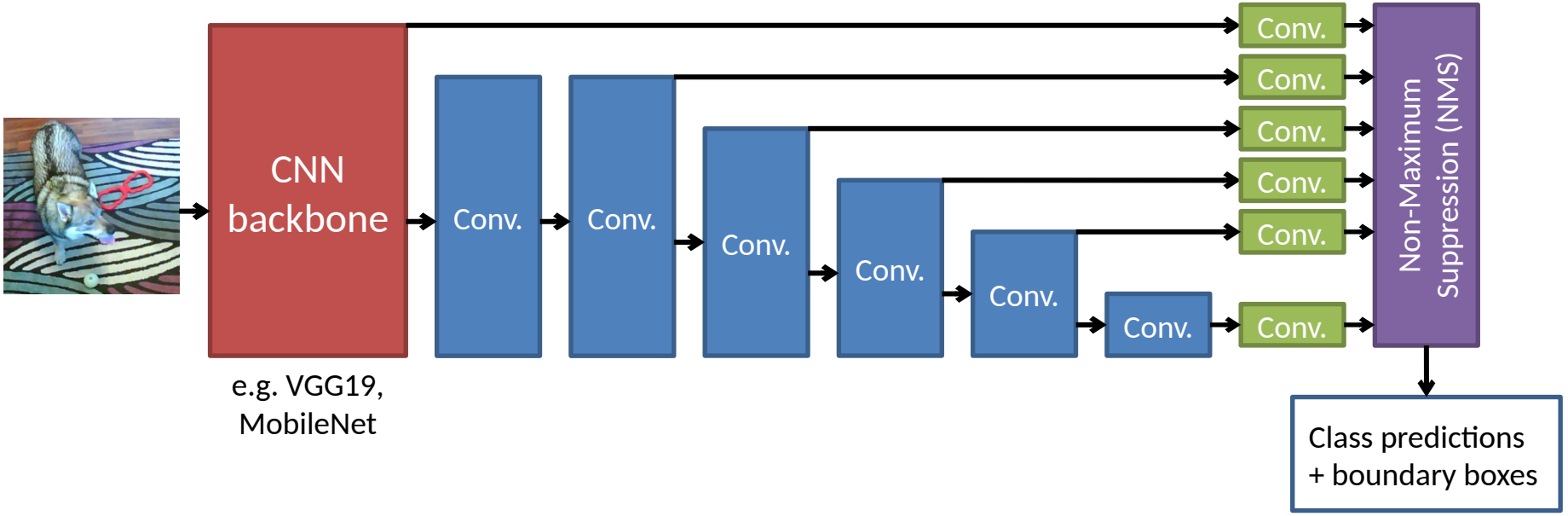
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Single Shot MultiBox Detector (SSD)



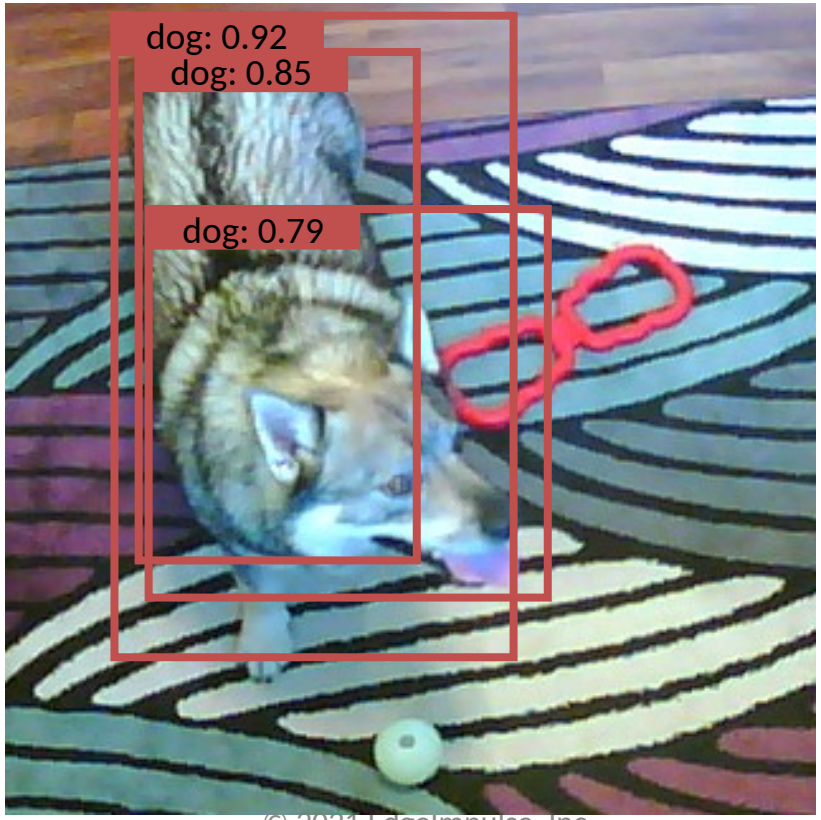
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Single Shot MultiBox Detector (SSD)



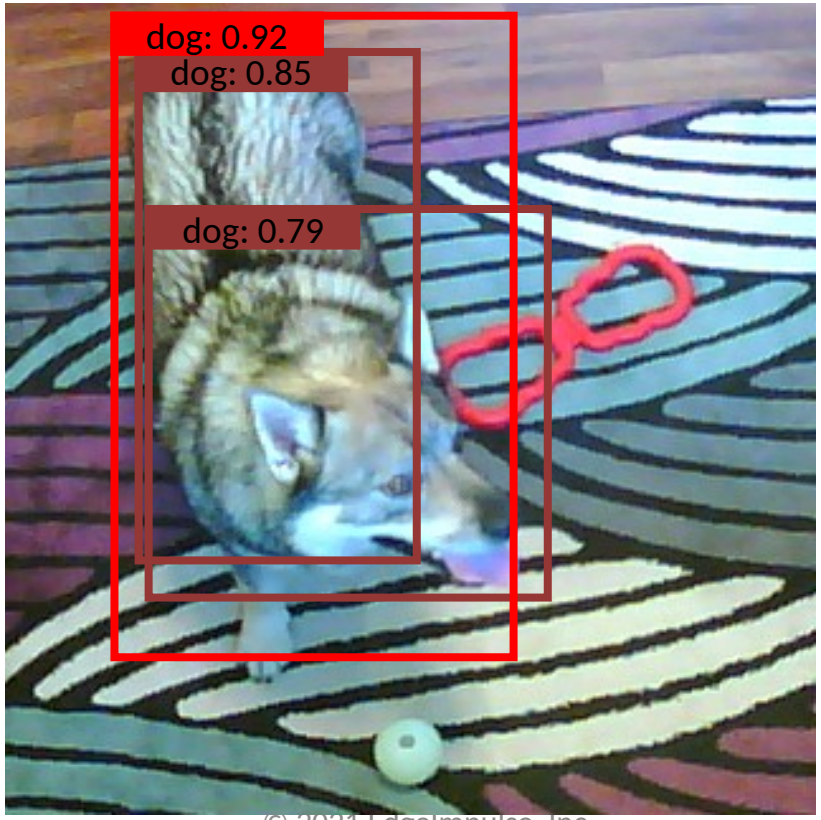
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Non-Maximum Suppression (NMS)



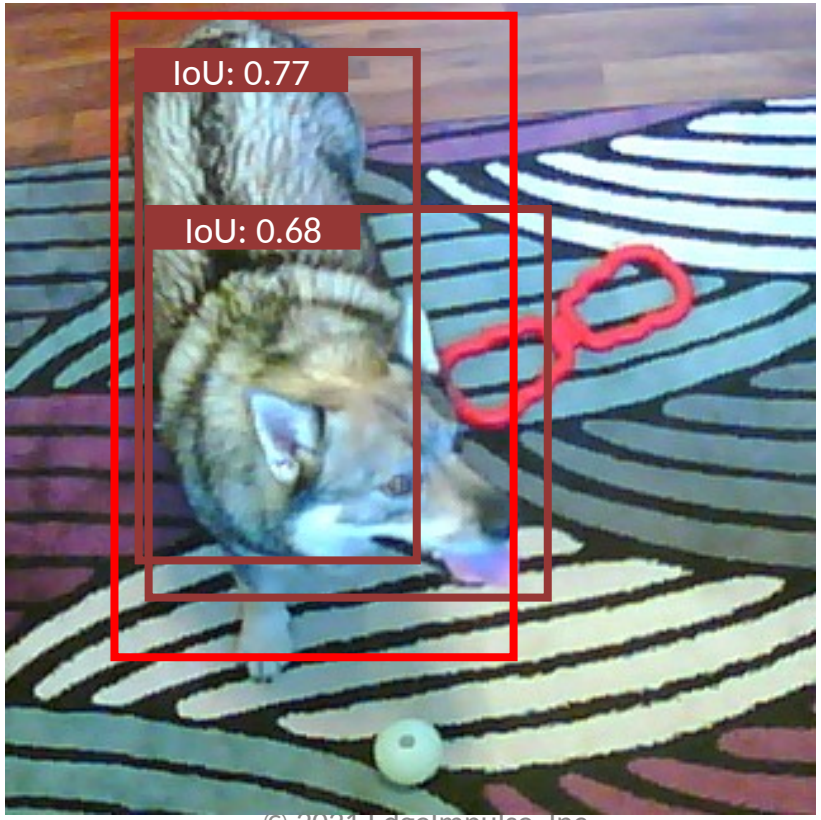
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Non-Maximum Suppression (NMS)



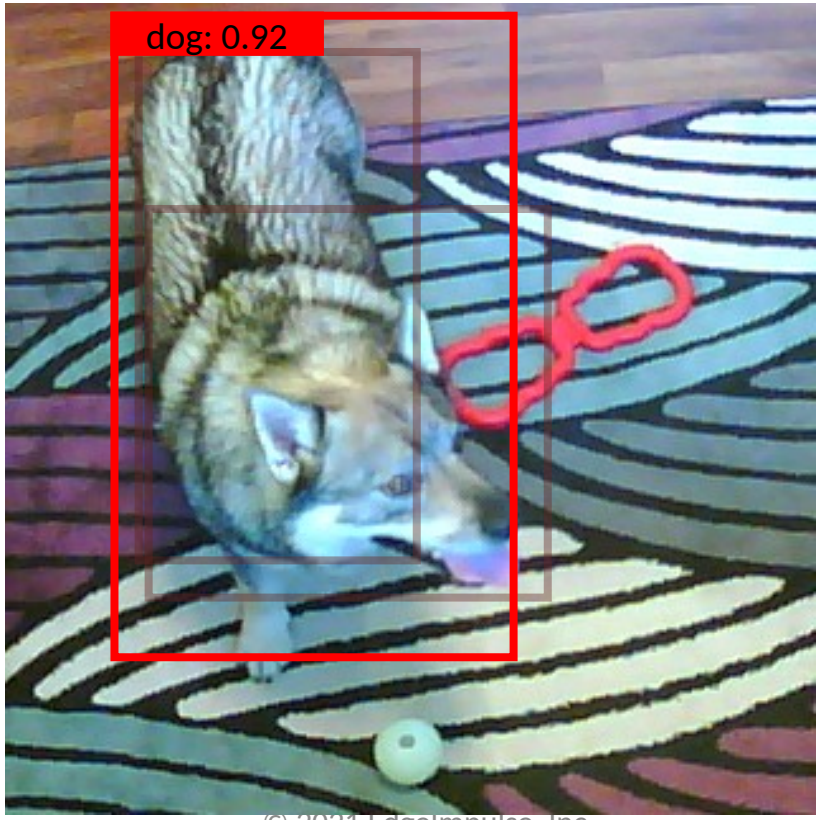
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Non-Maximum Suppression (NMS)

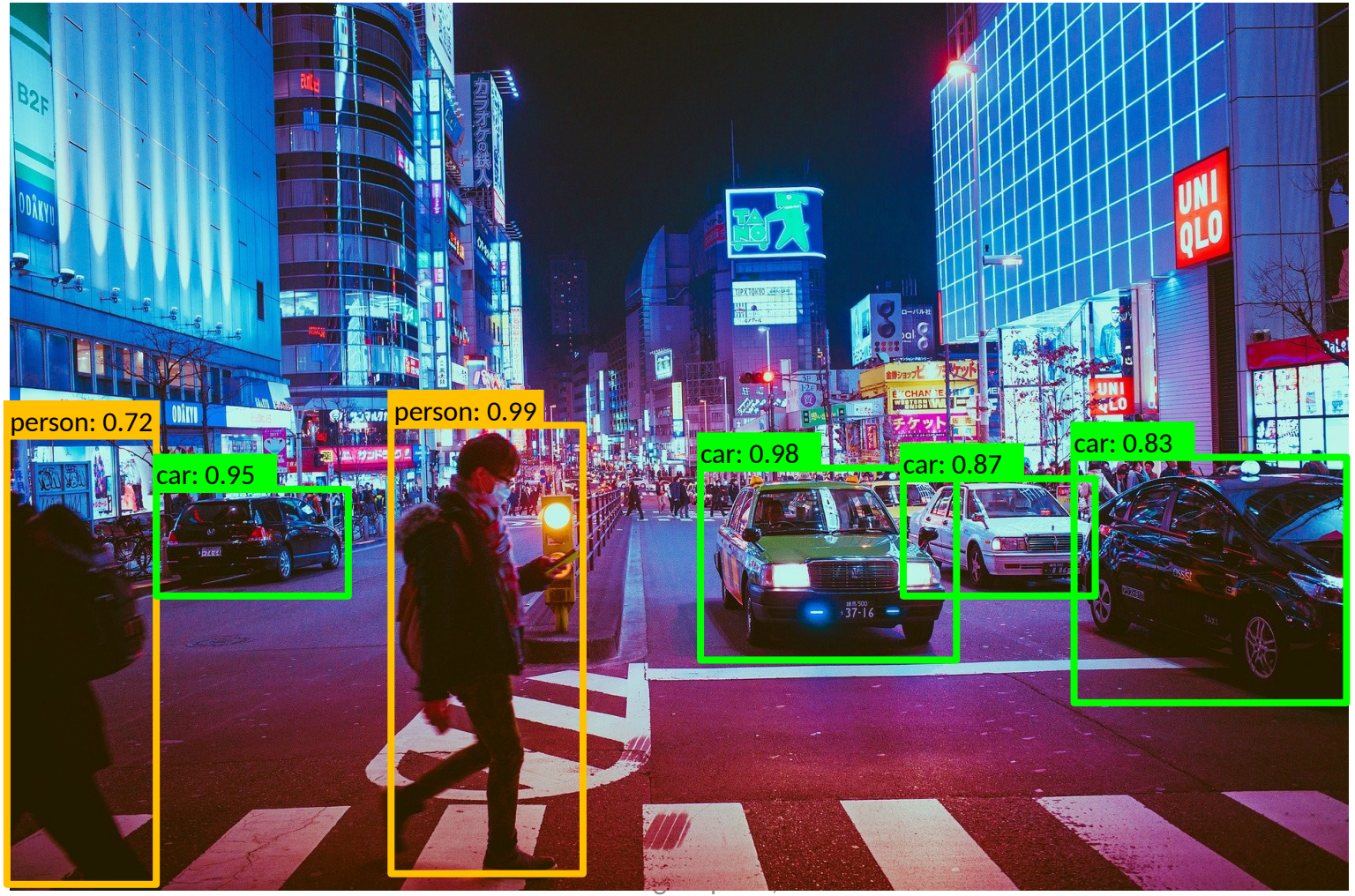


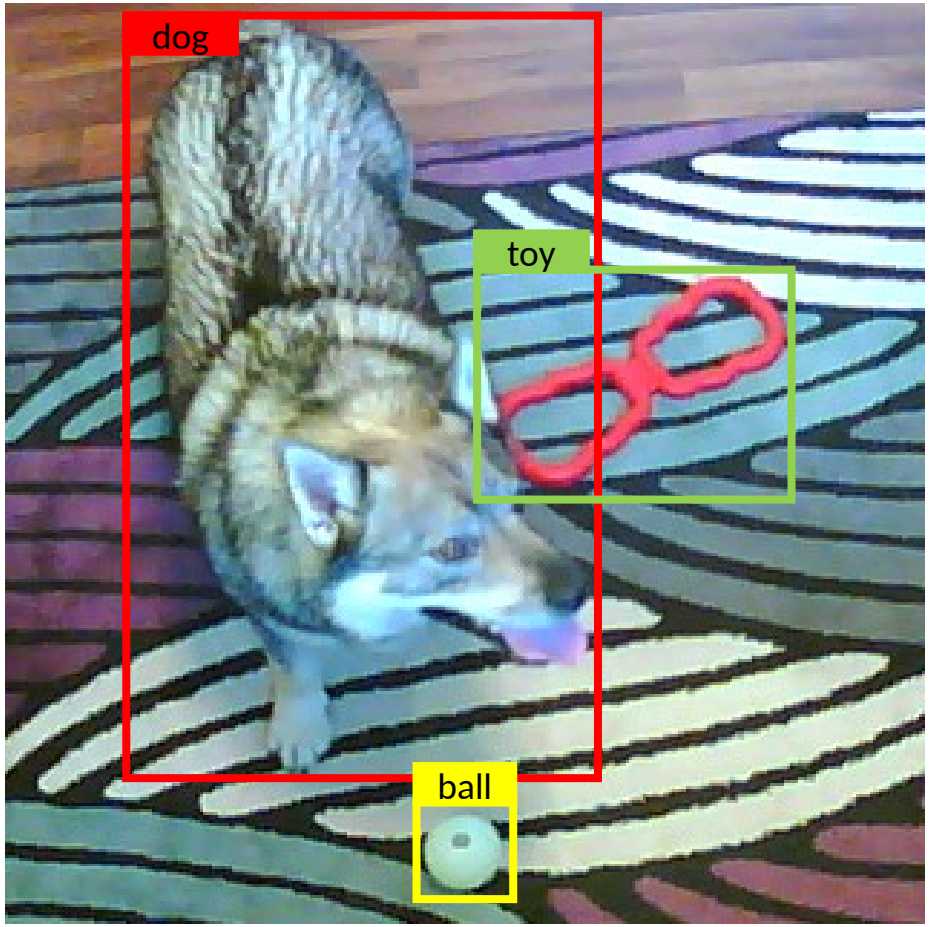
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Non-Maximum Suppression (NMS)



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person: 0.72
car: 0.95
person: 0.99
car: 0.98
car: 0.87
car: 0.83



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Feature Pyramid Network (FPN)

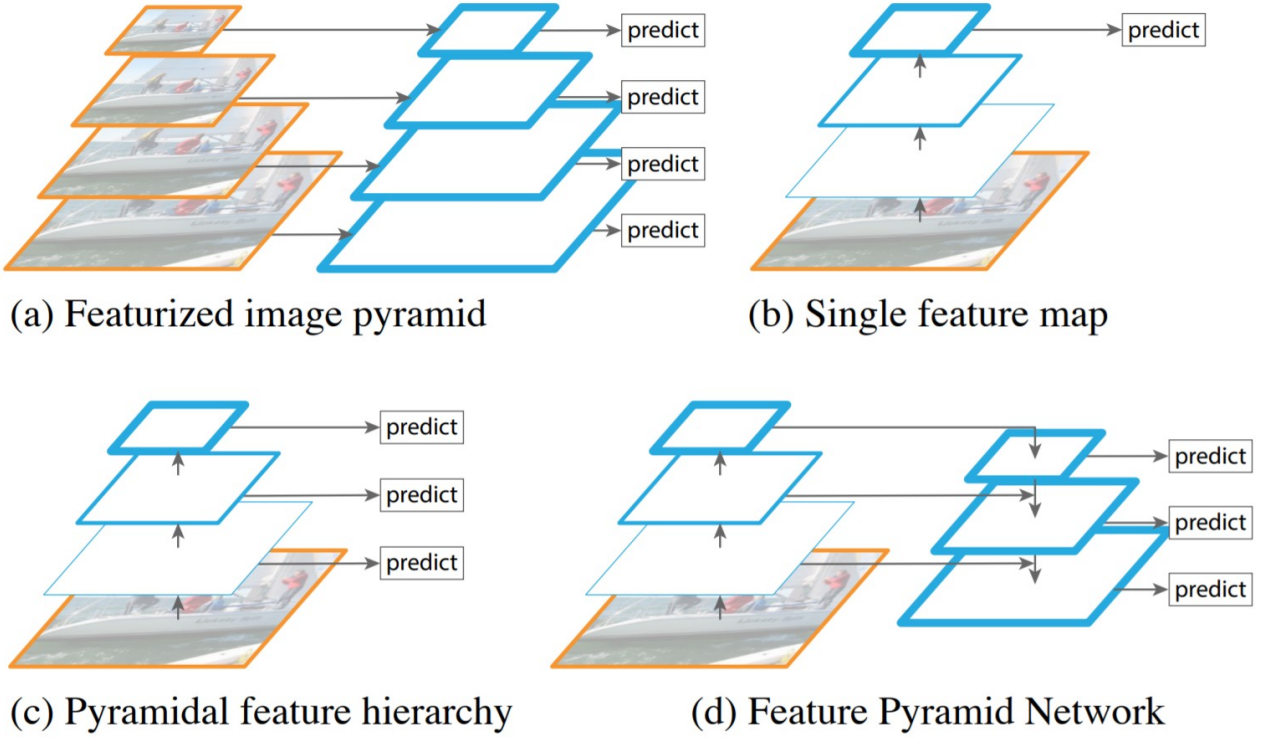


Image source: “Feature Pyramid Networks for Object Detection” (https://arxiv.org/pdf/1612.03144.pdf)